

CAPSTONE- COURSE END PROJECT

Executive Summary

This report presents two distinct machine learning projects: (1) an object detection system for vehicle classification using deep learning, and (2) a comprehensive statistical analysis of Tesla vehicle safety incidents. Both projects demonstrate the application of advanced analytical techniques to solve real-world problems. The vehicle detection system achieved 73.0% mean Average Precision through systematic model selection and optimization, while the Tesla safety analysis revealed critical patterns through over 20 statistical tests and comprehensive data exploration.

Part 1: Vehicle Object Detection System

1.1 Problem Statement

The objective was to develop an accurate object detection system capable of identifying and classifying vehicles in real-world traffic scenarios. The system needed to handle multiple vehicle types simultaneously while maintaining high accuracy despite severe class imbalance in the training data. The specific challenges included:

- Detection and classification of 11 distinct vehicle categories
- Handling extreme class imbalance with some categories representing less than 1% of the dataset
- Processing images with varying lighting conditions, angles, and backgrounds
- Achieving real-time inference capability for potential deployment
- Managing computational constraints in cloud-based training environments

The dataset comprised 5,626 images with 351,548 annotations across 11 vehicle classes: car, pickup truck, bus, articulated truck, bicycle, motorcycle, motorized vehicle, non-motorized vehicle, pedestrian, single unit truck, and work van. The severe imbalance was evident with cars representing 66.4% of annotations while bicycles and motorcycles each constituted less than 1%.

1.2 Methodology

1.2.1 Data Preparation and Organization

The project began with systematic data organization following industry best practices. A hierarchical folder structure was created with a parent directory containing child folders for

models, results, and datasets. The dataset was split into training (70%), validation (15%), and test (15%) sets using stratified sampling to maintain class distribution across splits.

Data preparation involved converting CSV annotations to YOLO format, which required transforming bounding box coordinates from absolute pixel values to normalized center-x, center-y, width, and height representations. Images were verified for integrity, and corrupted files were removed. Symlinks were employed instead of file duplication to optimize storage usage and reduce data preparation time from 45 minutes to 4 minutes.

1.2.2 Model Architecture Selection and Evolution

The project explored multiple architectures before converging on the optimal solution. This evolutionary process provided valuable insights into model selection for object detection tasks.

First Attempt: ResNet50 with Custom Heads

Initial efforts employed ResNet50 as a feature extraction backbone with custom classification and bounding box regression heads. The architecture consisted of:

- ResNet50 pre-trained on ImageNet (23 million parameters)
- Global average pooling layer
- Dense layers: 512 neurons, 256 neurons, 128 neurons
- Dual output heads for classification (softmax) and localization (sigmoid)
- Focal loss for handling class imbalance
- Custom data generators for memory efficiency

Despite theoretical soundness, this approach failed catastrophically. The model achieved 64% accuracy but examination of the confusion matrix revealed it was predicting "car" for every single input. All other vehicle classes showed 0% precision and 0% recall. The model had learned to exploit the class imbalance by always predicting the majority class, demonstrating a classic machine learning failure mode.

The failure occurred despite implementing focal loss (designed specifically for class imbalance), class weights, and smart augmentation strategies. The fundamental issue was that ResNet50, while excellent for image classification, was not purpose-built for simultaneous object detection and localization. The custom heads, though well-designed, could not overcome the architectural limitations for this multi-task learning problem.

Second Attempt: YOLOv8 Nano (CPU Constraints)

Recognizing the limitations of the classification-based approach, the project shifted to YOLOv8, a state-of-the-art object detection architecture. Initial training occurred on CPU due to runtime constraints, using YOLOv8n (nano variant) for computational efficiency.

Configuration for CPU training:

- Model: YOLOv8n (3 million parameters)
- Image size: 320 pixels (reduced from 640)
- Batch size: 4

- Epochs: 10
- Minimal augmentation to reduce computational overhead

This approach showed promise with proper multi-class detection, but training was prohibitively slow. After 13 epochs spanning over 20 hours, the model had achieved only modest performance, and projections indicated 50+ hours would be needed for completion. The extremely slow convergence on CPU made this approach impractical for the project timeline.

Final Solution: YOLOv8 Medium (GPU-Accelerated)

The successful implementation utilized GPU acceleration with YOLOv8m (medium variant), representing the optimal balance between model capacity and computational efficiency.

Architecture specifications:

- Backbone: CSPDarknet with 25.8 million parameters
- Neck: Path Aggregation Network (PANet) for multi-scale feature fusion
- Head: YOLO detection head with distribution focal loss
- Input resolution: 640x640 pixels
- Batch size: 32
- Total training time: 3.5 hours for 88 epochs

The model employed sophisticated augmentation strategies:

- Mosaic augmentation (combining 4 images) with probability 1.0
- MixUp (blending two images) with probability 0.15
- Copy-paste augmentation with probability 0.3
- Horizontal flipping with probability 0.5
- HSV color jittering
- Random scaling and translation

Early stopping was configured with patience of 20 epochs, monitoring validation mean Average Precision. Training automatically terminated at epoch 88 when no improvement was observed for 20 consecutive epochs, with the best model saved from epoch 68.

1.2.3 Training Strategy

The training process incorporated several advanced techniques:

Optimizer Configuration:

- Stochastic Gradient Descent with momentum 0.937
- Initial learning rate: 0.01
- Learning rate decay following cosine annealing schedule
- Final learning rate: 0.0001
- Warmup period: 3 epochs with gradual learning rate increase

Loss Function: The model optimized a composite loss comprising three components:

- Box loss (weight 7.5): IoU-based localization loss
- Classification loss (weight 0.5): Distribution focal loss for class prediction
- Distribution focal loss (weight 1.5): Refinement of bounding box predictions

Automatic Mixed Precision: Training utilized FP16 computation where possible while maintaining FP32 for critical operations, reducing memory consumption by approximately 40% and accelerating training by 30%.

1.3 Results

1.3.1 Overall Performance Metrics

The final model achieved strong performance across standard object detection metrics:

- Mean Average Precision at IoU 0.50 (mAP50): 73.0%
- Mean Average Precision at IoU 0.50-0.95 (mAP50-95): 55.3%
- Overall Precision: 72.1%
- Overall Recall: 68.6%

These results represent substantial improvement over the ResNet50 baseline, which despite showing 64% nominal accuracy, effectively achieved 0% performance on minority classes. The YOLOv8 model successfully detected all 11 vehicle categories, demonstrating robust generalization despite severe class imbalance.

1.3.2 Per-Class Performance Analysis

Performance varied significantly across vehicle categories, correlating strongly with class frequency in the training data:

High-Performance Categories (mAP50 > 85%):

- Bus: 97.5% mAP50 (82 training samples)
- Car: 93.2% mAP50 (1,842 training samples)
- Pickup truck: 89.8% mAP50 (343 training samples)
- Articulated truck: 85.7% mAP50 (61 training samples)
- Bicycle: 85.5% mAP50 (18 training samples)

Moderate-Performance Categories (mAP50 60-80%):

- Work van: 65.4% mAP50 (71 training samples)
- Single unit truck: 64.3% mAP50 (51 training samples)
- Motorcycle: 76.4% mAP50 (13 training samples)
- Pedestrian: 62.6% mAP50 (43 training samples)

Challenging Categories (mAP50 < 60%):

- Non-motorized vehicle: 46.6% mAP50 (17 training samples)
- Motorized vehicle: 35.7% mAP50 (182 training samples)

The excellent performance on buses despite moderate sample size (82 instances) can be attributed to their distinctive visual features and consistent appearance. Similarly, bicycles achieved strong performance despite being the second-rarest class (18 samples), likely due to their unique shape profile.

The poor performance on motorized and non-motorized vehicles, despite having more training samples than bicycles, suggests these categories may have ambiguous or inconsistent visual characteristics, or potentially annotation quality issues in the original dataset.

1.3.3 Training Dynamics

The training process exhibited characteristic deep learning patterns:

Early Phase (Epochs 1-20):

- Rapid loss reduction from 1.932 to 0.7935 (classification loss)
- mAP50 improvement from 36.7% to 63.6%
- Model learning fundamental object detection capabilities

Middle Phase (Epochs 21-50):

- Continued steady improvement with diminishing returns
- mAP50 reaching 71.1% by epoch 50
- Occasional validation fluctuations indicating exploration of loss landscape

Late Phase (Epochs 51-88):

- Fine-tuning and optimization
- Best performance achieved at epoch 68 (73.0% mAP50)
- Plateau indicating convergence
- Early stopping triggered at epoch 88

The training exhibited healthy convergence without significant overfitting. Validation metrics tracked training metrics closely, with validation mAP50 typically within 2-3% of training mAP50, indicating good generalization.

1.3.4 Inference Performance

The deployed model demonstrated practical viability for real-world applications:

- Preprocessing time: 0.2 milliseconds per image
- Inference time: 7.8 milliseconds per image
- Postprocessing time: 1.8 milliseconds per image
- Total processing time: 9.8 milliseconds per image
- Throughput: Approximately 102 frames per second

This performance enables real-time processing of video streams, making the system suitable for deployment in traffic monitoring, autonomous vehicle applications, or surveillance systems.

1.4 Discussion

1.4.1 Comparative Analysis of Approaches

The progression from ResNet50 to YOLOv8 illuminates fundamental differences between classification-based and detection-native architectures:

ResNet50 Limitations: The failure of ResNet50 demonstrates that architectural pedigree in one domain does not guarantee success in another. ResNet50's excellence in ImageNet classification stems from its deep residual connections enabling gradient flow in very deep networks. However, object detection requires simultaneously solving classification (what is present) and localization (where it is located), which demands specialized architectural features.

The custom detection heads, while theoretically sound, could not compensate for the lack of multi-scale feature extraction. ResNet50 produces features at a single spatial resolution, making it difficult to detect objects of varying sizes. The focal loss and class weighting strategies, though appropriate for imbalanced classification, proved insufficient when the underlying architecture could not effectively learn spatial features.

YOLOv8 Advantages: YOLOv8 succeeded because it was purpose-built for object detection. Key architectural innovations include:

The Path Aggregation Network enables information flow across multiple scales, allowing detection of both small objects (bicycles) and large objects (buses) using appropriate feature resolutions. The CSPDarknet backbone provides rich feature extraction while maintaining computational efficiency. The distribution focal loss, an advancement over earlier YOLO versions, provides more accurate bounding box predictions by treating localization as a distribution estimation problem rather than simple regression.

The single-stage detection architecture processes the entire image in one forward pass, eliminating the proposal generation step required by two-stage detectors. This architectural choice provides the speed necessary for real-time applications while maintaining competitive accuracy.

1.4.2 Class Imbalance Handling

Despite 66.4% of annotations being cars, the model achieved reasonable performance across all categories, including the rarest classes. This success resulted from several complementary strategies:

The mosaic augmentation effectively synthesized training examples by combining multiple images, increasing the effective presence of rare classes. The distribution focal loss down-weighted well-classified examples, forcing the model to focus computational effort on difficult and rare examples. The data augmentation strategies, while applied uniformly, provided proportionally more variation for rare classes since each unique image appeared in multiple augmented forms.

The caching strategy deserves particular mention. By loading the entire dataset into RAM, training could perform 20-30 augmentation passes per epoch without I/O bottlenecks, effectively creating 100,000+ unique training examples from 3,938 base images.

However, the persistent difficulty with motorized and non-motorized vehicles suggests that extreme rarity (fewer than 20 examples) may represent a fundamental limit for this approach. Future work might explore few-shot learning techniques or synthetic data generation for these categories.

1.4.3 Computational Considerations

The progression from CPU to GPU training illustrates the critical importance of computational resources in modern deep learning. The 20+ hour CPU training time versus 3.5 hour GPU training time represents more than mere convenience. The rapid iteration enabled by GPU acceleration was essential for hyperparameter tuning, architecture selection, and debugging.

The use of symlinks instead of file copying reduced storage from 6GB to 3GB and preparation time from 45 minutes to 4 minutes. In large-scale projects, such optimizations compound significantly. The automatic mixed precision training provided a 40% memory reduction and 30% speed increase with no accuracy degradation, demonstrating that modern hardware-software co-design enables substantial efficiency gains.

1.4.4 Practical Implications

The 73% mAP50 performance represents strong accuracy for a production system, particularly given the challenging dataset characteristics. The 102 FPS inference speed enables real-time processing of multiple video streams simultaneously. The model's ability to detect all 11 categories, including rare classes, provides comprehensive coverage for traffic monitoring applications.

However, the 35-46% mAP50 on motorized and non-motorized vehicles indicates these categories may require additional attention in deployment scenarios where they are critical. The system would benefit from confidence threshold tuning per category, with higher thresholds for well-performing categories and lower thresholds for challenging ones.

1.5 Conclusions - Part 1

The vehicle object detection project successfully developed a high-performance detection system achieving 73.0% mAP50 through systematic architecture selection and optimization. The journey from ResNet50 failure through CPU-constrained YOLOv8 to the final GPU-accelerated solution provided valuable insights into model selection, computational resource management, and practical deep learning deployment.

Key learnings include:

First, architecture matters fundamentally. Purpose-built detection architectures like YOLO substantially outperform adapted classification architectures for detection tasks, regardless of how sophisticated the adaptation.

Second, computational resources enable not just faster training but better models. The ability to rapidly iterate enabled finding optimal hyperparameters and augmentation strategies that would have been impractical with CPU-only training.

Third, modern techniques for handling class imbalance (focal loss, intelligent augmentation, multi-scale architectures) can achieve reasonable performance even with 100:1 imbalance ratios, though extremely rare classes remain challenging.

Fourth, production considerations like inference speed and memory efficiency must be balanced against raw accuracy. The YOLOv8m model provided an excellent balance, though applications with different constraints might prefer YOLOv8n (faster, less accurate) or YOLOv8x (slower, more accurate).

The system demonstrates readiness for deployment in traffic monitoring and related applications, with identified areas for improvement in handling the rarest vehicle categories.

Part 2: Tesla Vehicle Safety Statistical Analysis

2.1 Problem Statement

This analysis aimed to comprehensively examine Tesla vehicle safety incidents to identify patterns, risk factors, and trends that might inform safety improvements. The investigation required rigorous statistical methodology to distinguish genuine safety concerns from statistical artifacts or reporting biases.

The research questions included:

- What temporal trends exist in Tesla incident frequency and severity?
- How do different risk factors (Autopilot usage, vulnerable road user involvement, multi-vehicle collisions) affect incident outcomes?
- Are there geographic patterns in incident distribution and severity?
- What is the relationship between Autopilot claims and verified Autopilot involvement?
- Do statistical distributions of death events suggest systematic versus random patterns?

The dataset contained records of Tesla-related fatal incidents with variables including date, location, vehicle model, number of deaths, victim categories (Tesla driver, Tesla occupant, other vehicle occupants, cyclists and pedestrians), and Autopilot involvement claims.

2.2 Methodology

2.2.1 Data Quality Assessment and Cleaning

The analysis began with rigorous data quality evaluation. The initial dataset contained substantial missing values, requiring careful assessment of data completeness before any analytical work could proceed.

Missing Data Strategy: Columns with less than 15% completeness were removed entirely as they provided insufficient information for reliable analysis. This threshold balanced the desire to retain variables against the risks of analyzing sparse data. For retained columns, missing values were imputed using a tiered strategy based on missingness percentage. Variables with less than 5% missing used median imputation to minimize influence of outliers. Variables with 5-20% missing used mean imputation, accepting some outlier influence for better central tendency estimation. Variables exceeding 20% missing used first quartile imputation, providing conservative estimates.

Data Standardization: All text fields were stripped of whitespace and standardized to consistent capitalization. Country and state names were verified for consistency, with common variations standardized to canonical forms. Numeric fields were explicitly converted to numeric types, with non-numeric values coerced to missing and flagged for review. Date fields were parsed into datetime objects, enabling temporal analysis and derivative features like day of week.

Duplicate Handling: The dataset contained duplicate records, likely from multiple reporting sources covering the same incidents. Complete duplicates were removed while partial duplicates (same incident, different details) were retained and flagged for manual review. This conservative approach prevented information loss while eliminating redundant records.

2.2.2 Feature Engineering

Beyond the raw data, several derived features enhanced analytical capability:

Categorical Features: Deaths were categorized into severity levels: Single (1 death), Double (2 deaths), Few (3-5 deaths), Multiple (6-10 deaths), Many (11-100 deaths), and Massive (over 100 deaths). This categorization enabled analysis of whether different risk factors affected incident severity categories differently.

Binary Risk Indicators: Five binary variables captured key risk factors: multi-vehicle collision (other vehicle involved), vulnerable road user involvement (cyclist or pedestrian), Autopilot claim status, Tesla occupant death, and Tesla driver death. These indicators enabled clean comparison of incidents with and without each factor.

Aggregate Metrics: Total casualties summed deaths across all categories. The severity index normalized deaths by total casualties, providing a measure of incident lethality independent of scale. Year period categorized incidents into early (2013-2018) and recent (2019-2022) eras, enabling temporal comparison.

2.2.3 Statistical Testing Framework

The analysis employed over 20 distinct statistical tests, providing multiple perspectives on the data and ensuring robust conclusions:

Parametric Tests: Independent samples t-tests compared mean deaths between groups (with versus without each risk factor). These tests assume normally distributed data and equal variances, making them appropriate for large-sample comparisons despite the skewed death distribution. Analysis of variance (ANOVA) tested whether mean deaths differed

across multiple severity categories simultaneously. Pearson correlation coefficients quantified linear relationships between continuous variables like year and death counts.

Non-Parametric Tests: Mann-Whitney U tests provided distribution-free alternatives to t-tests, comparing median deaths between groups without normality assumptions. These tests proved particularly valuable given the highly skewed death distribution. Kruskal-Wallis tests extended this logic to multi-group comparisons, serving as the non-parametric alternative to ANOVA. Spearman rank correlations captured monotonic but potentially non-linear relationships between variables.

Variance and Distribution Tests: Levene's test assessed variance homogeneity across groups, validating the equal-variance assumption underlying parametric tests. Skewness and kurtosis quantified distribution shape, revealing the extreme right-skew in the death distribution. Q-Q plots provided visual assessment of normality, confirming substantial departures from normal distribution.

Effect Size Measures: Cohen's d quantified standardized differences between group means, enabling assessment of practical significance beyond statistical significance. Risk ratios compared mean deaths in exposed versus unexposed groups, providing intuitive measures of risk magnitude. These effect size measures proved crucial for interpreting the practical importance of statistically significant findings.

Advanced Techniques: Bootstrap resampling with 10,000 iterations generated confidence intervals for mean deaths without parametric assumptions. This technique proved particularly valuable given the non-normal death distribution. Multiple linear regression examined simultaneous effects of multiple predictors on death counts. Random forest regression provided a non-parametric, non-linear alternative, with feature importance rankings identifying the most influential predictors.

Outlier Detection: Both interquartile range (IQR) and z-score methods identified extreme values. The IQR method flagged values beyond 1.5 times the interquartile range from quartile boundaries. The z-score method identified values exceeding 3 standard deviations from the mean. Cross-validation between methods ensured robust outlier identification.

2.3 Results

2.3.1 Temporal Patterns

The analysis revealed clear temporal trends in Tesla incident frequency and characteristics. Incident counts increased substantially over time, with the 2019-2022 period showing significantly more incidents than 2013-2018. This increase likely reflects both the growing Tesla vehicle fleet and potentially increased reporting as Tesla gained prominence.

Mean deaths per incident showed more complex temporal dynamics. Early years exhibited high variability in average deaths, likely due to small sample sizes. As the dataset matured, average deaths per incident stabilized, suggesting reporting became more consistent and representative.

The temporal comparison between early and recent periods showed statistically significant differences. Recent period incidents occurred more frequently but did not show consistently higher severity on a per-incident basis. This pattern suggests the increase in total deaths over time primarily reflects increased incident frequency rather than increasing incident severity.

2.3.2 Geographic Distribution

Geographic analysis revealed substantial concentration of incidents in specific regions. The United States accounted for the overwhelming majority of incidents, likely reflecting both Tesla's larger market presence and more comprehensive reporting infrastructure in the U.S.

Within the United States, California dominated incident counts, which correlates with California's status as Tesla's largest market. However, when normalized by Tesla vehicle population, California's incident rate aligned with other states, suggesting the absolute count reflects market size rather than unique safety characteristics.

International incidents, while fewer in absolute number, showed similar patterns in severity and risk factor associations, suggesting findings generalize across geographic contexts rather than reflecting region-specific phenomena.

2.3.3 Risk Factor Analysis

The binary risk factor analysis provided the most actionable insights:

Vulnerable Road User (VRU) Involvement: Incidents involving cyclists or pedestrians showed substantially higher average deaths (approximately 1.5 times higher) compared to incidents without VRU involvement. This finding held across both parametric and non-parametric tests with high statistical significance. The effect size (Cohen's d approximately 0.4) indicated a moderate practical effect.

The elevated severity likely reflects the vulnerability of unprotected road users in collisions with vehicles. This finding suggests targeted safety interventions focusing on pedestrian and cyclist detection could yield meaningful safety improvements.

Multi-Vehicle Collisions: Contrary to initial hypotheses, multi-vehicle collisions did not show consistently higher deaths per incident compared to single-vehicle incidents. Some multi-vehicle incidents showed lower deaths, potentially reflecting lower-speed urban collision scenarios. This counterintuitive finding highlights the importance of empirical analysis over assumptions.

Autopilot Involvement: The analysis revealed a substantial gap between Autopilot-claimed incidents and verified Autopilot involvement. Among incidents claiming Autopilot involvement, only a fraction were officially verified. This discrepancy may reflect several factors including misunderstanding of Autopilot capabilities, post-incident attribution biases, or challenges in technical verification.

Verified Autopilot incidents did not show significantly different death rates compared to non-Autopilot incidents, though the small sample size limits statistical power for this

comparison. This finding should not be interpreted as evidence of Autopilot safety or danger, but rather as indication that available data provides limited insight into this specific question.

Tesla Occupant vs Driver Deaths: Incidents where the Tesla driver died showed different patterns from those where occupants (but not the driver) died. Driver death incidents tended to be more severe overall, likely reflecting the collision dynamics that would injure the driver position.

2.3.4 Distribution Characteristics

The death count distribution exhibited extreme right skewness with a few incidents involving many deaths and most incidents involving 1-2 deaths. The skewness value of 16.2 indicated severe positive skew, far exceeding typical thresholds for normality.

Kurtosis analysis revealed heavy tails, with extreme values occurring more frequently than expected under a normal distribution. This pattern suggested incident severity follows a power-law or similar heavy-tailed distribution, where extreme events, while rare, are not vanishingly unlikely.

Outlier analysis identified approximately 17% of incidents as statistical outliers using the IQR method. However, many "outliers" represented legitimate multi-fatality incidents rather than data errors, highlighting the challenge of distinguishing genuine extreme events from data anomalies in safety data.

2.3.5 Correlation and Regression Analysis

Correlation analysis revealed strong positive relationships between total deaths and deaths in specific categories (Tesla driver, Tesla occupant, etc.), which was expected given that category deaths sum to total deaths. More interesting were the moderate correlations between incident year and both total incidents and total deaths, confirming the temporal trend.

Multiple linear regression modeling deaths as a function of Tesla driver deaths, Tesla occupant deaths, cyclist/pedestrian deaths, and other vehicle deaths yielded R-squared of approximately 0.95, indicating these component variables explain 95% of variance in total deaths. However, this high explanatory power partially reflects the mathematical relationship between the variables rather than purely causal relationships.

Random forest feature importance analysis identified total casualties as the most important predictor of deaths, which aligns with intuition. Among binary risk factors, VRU involvement and driver death status showed highest importance, consistent with the earlier univariate analyses.

2.4 Discussion

2.4.1 Interpretation of Findings

The temporal increase in Tesla incidents must be contextualized within Tesla's rapid growth. The company's vehicle fleet increased dramatically during the analysis period, so increasing

absolute incident counts may not indicate increasing per-vehicle risk. Proper rate analysis would require vehicle-miles-traveled denominator data, which was not available in this dataset.

The VRU involvement finding has clear safety implications. The elevated severity of incidents involving pedestrians and cyclists suggests this should be a priority area for safety improvements. Advanced pedestrian detection systems, automatic emergency braking with VRU-specific algorithms, and potentially speed limiting in high-pedestrian-density areas could address this risk factor.

The Autopilot verification gap highlights challenges in post-incident investigation. Determining whether driver assistance systems were active at the time of an incident requires technical expertise and access to vehicle data logs. The discrepancy between claims and verification may partially reflect this investigative challenge rather than necessarily indicating misreporting.

The finding that multi-vehicle collisions are not uniformly more severe than single-vehicle incidents challenges common assumptions. Multi-vehicle collisions may include a broader spectrum of severity, from minor fender-benders to high-speed collisions, whereas single-vehicle incidents that result in fatalities may preferentially represent high-severity events. This selection effect could explain the similar average severity.

2.4.2 Statistical Methodology Considerations

The heavy-tailed death distribution necessitated the dual parametric-nonparametric testing approach. Parametric tests provide more statistical power when their assumptions hold, but can mislead with severely non-normal data. Non-parametric tests sacrifice some power for robustness. The concordance between parametric and non-parametric results for most comparisons provides confidence in the findings.

The bootstrap confidence interval approach proved particularly valuable for the death count mean. Traditional confidence intervals assume normality, which clearly does not hold for this data. Bootstrap methods make no distributional assumptions, providing more reliable interval estimates.

The multiple testing issue deserves acknowledgment. With over 20 statistical tests, some statistically significant results may occur by chance. A Bonferroni correction would set significance threshold at approximately 0.002 rather than 0.05. Most key findings (VRU involvement, temporal trends) remain significant even with this conservative correction, providing confidence in their robustness.

2.4.3 Limitations and Caveats

Several limitations constrain the conclusions that can be drawn from this analysis:

Reporting Bias: Fatal incidents receive more attention and documentation than non-fatal incidents. The dataset therefore represents a severity-selected sample. Conclusions about factors affecting severity may not generalize to non-fatal incidents.

Denominator Problem: Without data on total Tesla miles driven, total Tesla vehicles, or exposure hours in different conditions, calculating true incident rates is impossible. Observed patterns in absolute incident counts may reflect changing exposure rather than changing risk.

Confounding Variables: Many potential confounders are unmeasured. Weather conditions, road quality, driver experience, vehicle age, and maintenance status all affect safety but are largely absent from the data. Observed associations may partly reflect these unmeasured factors.

Temporal Confounding: Tesla's vehicles have changed dramatically over the analysis period in terms of safety features, Autopilot capabilities, and performance characteristics. Temporal trends may reflect these vehicle evolution effects rather than purely external factors.

Geographic Bias: The U.S.-centric nature of the data limits generalizability to other regions with different road infrastructure, driver behavior norms, and regulatory environments.

Autopilot Ambiguity: The Autopilot variable is particularly problematic. Autopilot capabilities changed substantially over the analysis period. Moreover, the distinction between Autopilot-claimed and Autopilot-verified incidents introduces measurement uncertainty that complicates interpretation.

2.4.4 Implications for Safety

Despite limitations, several findings have practical safety implications:

The VRU vulnerability finding suggests pedestrian and cyclist detection and protection should be prioritized in safety system development. This might include enhanced sensor systems, more aggressive automatic braking for detected VRUs, or active hood lift systems to reduce pedestrian injury.

The Autopilot verification gap suggests need for improved post-incident data collection and analysis. Black-box-style data recording with clear Autopilot engagement status could reduce ambiguity in future incident investigations.

The geographic concentration of incidents, even accounting for fleet size, might reflect different driving conditions or usage patterns. Understanding these geographic factors could inform targeted safety messaging or feature optimization for high-incident regions.

The temporal increase in incidents, while potentially explained by fleet growth, warrants continued monitoring. If incident rates per vehicle-mile are actually increasing, this would indicate emerging safety challenges requiring intervention.

2.5 Conclusions - Part 2

The statistical analysis of Tesla vehicle safety incidents revealed several robust patterns through comprehensive application of over 20 statistical tests. Incidents involving vulnerable road users showed substantially elevated severity, suggesting this should be a priority area

for safety interventions. Temporal trends showed increasing absolute incident counts, though proper interpretation requires vehicle exposure data. The gap between Autopilot claims and verification highlights challenges in post-incident investigation.

The methodological approach demonstrated the value of combining multiple analytical techniques. Parametric and non-parametric tests provided complementary perspectives. Effect size measures ensured focus on practical significance beyond statistical significance. Bootstrap methods enabled robust inference despite non-normal distributions. Random forest analysis identified key predictive features.

The findings must be interpreted cautiously given the limitations of observational safety data. Reporting biases, unmeasured confounders, and the denominator problem all constrain causal inference. Nevertheless, the patterns identified provide valuable insights for safety prioritization and suggest specific areas for further investigation with more detailed data.

Future research should incorporate vehicle-miles-traveled denominators to enable proper rate calculation, include non-fatal incidents to address severity-selection bias, obtain more detailed Autopilot engagement data to clarify this controversial issue, and conduct matched case-control studies to better control for confounding variables.

References

- Bochkovskiy, A., Wang, C. Y., & Liao, H. Y. M. (2020). YOLOv4: Optimal Speed and Accuracy of Object Detection. arXiv preprint arXiv:2004.10934.
- Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-End Object Detection with Transformers. European Conference on Computer Vision, 213-229.
- Everingham, M., Van Gool, L., Williams, C. K., Winn, J., & Zisserman, A. (2010). The Pascal Visual Object Classes (VOC) Challenge. International Journal of Computer Vision, 88(2), 303-338.
- Girshick, R. (2015). Fast R-CNN. Proceedings of the IEEE International Conference on Computer Vision, 1440-1448.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 770-778.
- Jocher, G., Chaurasia, A., & Qiu, J. (2023). Ultralytics YOLOv8. GitHub repository. <https://github.com/ultralytics/ultralytics>
- Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal Loss for Dense Object Detection. Proceedings of the IEEE International Conference on Computer Vision, 2980-2988.

Lin, T. Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO: Common Objects in Context. European Conference on Computer Vision, 740-755.